Lesson 5: Jump!

Module 2 | 45 minutes | Jumping with velocity | Grades 3–5 · ages 9–11

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

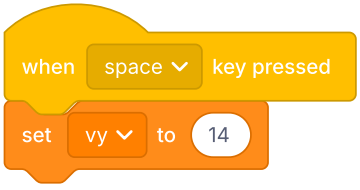
a. impulse ___ A. The curved path of a jump.
b. grounded ___ B. Standing on the floor.
c. arc ___ C. A single push that changes velocity.

2. Plan your code

Before you build it: Jump: gravity is already coded. When the space key is pressed, if the Hero is on the floor (y position < -118), set vy to 14. The Hero should jump up and land again. Write or draw the blocks you will use.

3. Debug the flying frog

The player can jump again and again in mid-air, flying off the top. Which script fixes it? Circle one.



a. A. A (script A)
b. B. B (script B)
c. C. C (script C)

4. Anatomy of a jump

Put a jump in order (what happens to vy) Number them 1–5.

a. ___ vy goes negative and the sprite falls
b. ___ vy reaches 0 at the top of the jump
c. ___ The sprite rises; gravity makes vy smaller each frame
d. ___ Space pressed: vy is set to 14
e. ___ The sprite lands and vy is set to 0

5. Plan your code

Before you build it: Walk and jump: add smooth walking (arrow keys, change x by 6) to the jumping Hero, so it can jump while moving. Write or draw the blocks you will use.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can make a sprite jump by setting its velocity.
☐ I can stop mid-air jumps.
☐ I can tune jump height.